

LEVEL 1 — BASIC Database & Table Fundamentals

Q1. Database + Basic Table

Create a database named college and select it. Then create a table student with the columns below.

Field	Requirement
student_id	INT, Primary Key, Auto Increment
name	VARCHAR(50), NOT NULL
age	INT
address	VARCHAR(100)

Q2. Data Types

Create a table employee using appropriate data types for the fields below. You must use INT, VARCHAR, DECIMAL, DATE, ENUM, BOOLEAN, and TIMESTAMP appropriately.

Field	Requirement
Employee ID	Unique identifier — INT
Employee Name	Text — VARCHAR
Salary	Precise money value — DECIMAL
Date of Birth	Calendar date — DATE
Gender	Male, Female, Other — ENUM
Is Active	TRUE / FALSE — BOOLEAN
Created Date & Time	Auto timestamp — TIMESTAMP

Q3. Constraints

Create a table product with the following rules. Give meaningful constraint names wherever applicable.

Field	Requirement
product_id	Primary Key + Auto Increment
name	NOT NULL
code	UNIQUE
price	DECIMAL(10,2), must be greater than 0
category	ENUM: Electronics, Clothing, Food
status	DEFAULT 'Available'

LEVEL 2 — ALTER TABLE Adding, Modifying & Renaming

Q4. Adding and Modifying Columns

Create a student table with basic columns, then perform the following ALTER TABLE operations in order:

- Add email as VARCHAR(100).
- Change email to VARCHAR(150).
- Rename email to student_email.
- Add phone as CHAR(10).
- Remove phone.
- Rename the table from student to students.

Q5. Adding Constraints

Create the following two tables:

Department	Employee
dept_id	emp_id
dept_name	name
	email
	dept_id

Then use ALTER TABLE to:

- Add Primary Key to dept_id.
- Add Primary Key to emp_id.
- Add UNIQUE constraint to email.
- Add a FOREIGN KEY on dept_id.
- Add a CHECK constraint for emp_id > 0.

Use your own constraint names.

Q6. NOT NULL, DEFAULT & AUTO_INCREMENT

Create a table student without using these three properties initially. Then use ALTER TABLE to:

- Make name NOT NULL.
 - Add AUTO_INCREMENT to student_id.
 - Add a DEFAULT value 'Kathmandu' to address.
 - Change the default value to 'Pokhara'.
 - Remove the DEFAULT value.
 - Remove NOT NULL from name.
 - Remove AUTO_INCREMENT from student_id.
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LEVEL 3 — REMOVING CONSTRAINTS Dropping What You Defined

Q7. Drop Constraints

Create a table containing a Primary Key, a Unique constraint, a Foreign Key, and a Check constraint — with meaningful constraint names. Then write queries to:

- Drop the Primary Key.
- Drop the Unique constraint.
- Drop the Foreign Key.
- Drop the Check constraint.

♦ *Do not delete any columns or records — only the constraints themselves.*

Q8. Constraint Investigation

Create a table containing at least: Primary Key, Foreign Key, UNIQUE, CHECK, and AUTO_INCREMENT. Then:

- Use SHOW CREATE TABLE to find the constraint names.
- Identify which names were explicitly defined by you.
- Identify automatically generated names.
- Remove the Foreign Key using its constraint name.
- Remove the CHECK constraint using its constraint name.

LEVEL 4 — DATABASE / TABLE MANAGEMENT

Inspecting Structure

Q9. Information Commands

Create a database containing at least two tables. Write queries to:

- Display all databases.
- Select your database.
- Display all tables.
- Display the structure of one table using DESC.
- Display the complete definition using SHOW CREATE TABLE.

♦ *Explain, in one or two sentences, the difference between DESC and SHOW CREATE TABLE.*

Q10. DROP vs TRUNCATE

Create a table named test_student. Then write queries demonstrating TRUNCATE TABLE and DROP TABLE. Before executing each, explain:

- What happens to the data?
- Does the table still exist?
- Can you continue using the table afterward?

♦ *Don't actually drop your main practice database.*

LEVEL 5 — COMPREHENSIVE Complete College Database

Q11. Complete College Database

Create a database called college_management with the following tables:

Department
dept_id
dept_name
floor

Student
student_id
name
email
gender
age
address
dept_id
created_at

Requirements — apply all of the following across the two tables:

- Appropriate data types
- Primary Keys
- Auto Increment
- NOT NULL
- UNIQUE
- ENUM
- CHECK
- DEFAULT
- TIMESTAMP
- Foreign Key
- Meaningful constraint names

Then perform at least 5 different ALTER TABLE operations on the tables.

LEVEL 6 — EXAM CHALLENGE Full Unit 5 Challenge

Q12. Full Unit 5 Challenge — Employee Table

You are given the following requirements for an employee table:

Field	Requirement
emp_id	Integer, Primary Key, Auto Increment
name	Variable text, maximum 50 characters, NOT NULL
email	Variable text, UNIQUE
salary	Exact decimal with 2 decimal places, greater than 0
gender	Only Male / Female / Other
dob	Date
is_active	Boolean, default TRUE
joined_at	Date and time
address	Variable text, default Kathmandu
dept_id	Foreign Key

Your tasks:

- Create the database.
- Create the table.
- Give meaningful names to applicable constraints.
- Verify the table using DESC.
- Verify the complete definition using SHOW CREATE TABLE.
- Add one new column using ALTER TABLE.
- Modify one existing column.
- Rename one column.
- Add one new constraint.
- Remove one constraint.
- Remove the newly added column.
- Finally, demonstrate the difference between TRUNCATE TABLE and DROP TABLE.